

ON EIGENFUNCTIONS OF BLOCK GRAPHS OF STEINER SYSTEMS

Dmitry Panasenko
makare95@mail.ru

Based on joint work with Sergey Goryainov

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Strongly regular graphs

A k -regular graph on v vertices is called **strongly regular** (SRG for short) with parameters (v, k, λ, μ) if any two adjacent vertices have λ common neighbours and any two distinct non-adjacent vertices have μ common neighbours.

A strongly regular graph G is **primitive** if both G and its complement are connected. If G is not primitive, we call it **imprimitive**. The imprimitive strongly regular graphs are exactly the disjoint unions of complete graphs and their complements.

Spectra of strongly regular graphs

Lemma 1 ([1, Theorem 5.2.1])

If G is a primitive strongly regular graph with parameters (v, k, λ, μ) and

$$\Delta := \sqrt{(\lambda - \mu)^2 + 4(k - \mu)},$$

then G has exactly three eigenvalues

$$k, \quad r = \frac{\lambda - \mu + \Delta}{2}, \quad s = \frac{\lambda - \mu - \Delta}{2} \quad (s < 0 < r),$$

with respective multiplicities

$$m_k = 1, \quad m_r = -\frac{(v-1)s + k}{r-s}, \quad m_s = \frac{(v-1)r + k}{r-s}.$$

Steiner systems

A **Steiner system** with parameters t, m, n , written $S(t, m, n)$, is a type of block design, where points are elements of set S of size n and blocks are m -element subsets of S with the property that each t -element subset of S is contained in exactly one block.

A Steiner system $S(2, m, n)$ is also known as a **$2-(n, m, 1)$ design** or just a **2-design**. The **block graph** of a $2-(n, m, 1)$ design is the graph with the blocks of the design as the vertices in which two blocks are adjacent if and only if they intersect.

A Steiner system $S(2, 3, n)$ is called a **Steiner triple system** $STS(n)$, and its blocks are called triples.

Block graphs of $2-(n, m, 1)$ designs

Theorem 1 ([1, Theorem 5.3.1])

The block graph of a $2-(n, m, 1)$ design (that is not symmetric) is strongly regular with parameters

$$v = \frac{n(n-1)}{m(m-1)}, k = \frac{m(n-m)}{m-1}, \lambda = (m-1)^2 + \frac{n-1}{m-1} - 2, \mu = m^2$$

and spectrum

$$\{k^1, k - n^{n-1}, -m^{v-n}\}.$$

[1] C. Godsil, K. Meagher, *Erdős-Ko-Rado Theorems: Algebraic Approaches*, Cambridge University Press (2015)

Affine spaces

Let $V(n, q)$ be a vector space of dimension n over finite field $\mathbb{F}_q$. The **affine space** $AG(n, q)$ is the geometry whose points, lines, planes, etc., are the cosets of the subspaces of $V(n, q)$ of dimension $0, 1, \dots, n$.

The affine space $AG(n, q)$ is a Steiner system $S(2, q, q^n)$, where the blocks are the lines of $AG(n, q)$. Let us call such a system an **affine Steiner system** and denote such a system $AS(n, q)$.

The **projective space** $PG(n, q)$ is the geometry whose points, lines, planes, etc., are the subspaces of $V(n + 1, q)$ of dimension $1, 2, \dots, n + 1$.

The projective space $PG(n, q)$ is a Steiner system $S(2, q + 1, (q^n - 1)/(q - 1))$, where the blocks are the lines of $PG(n, q)$. Let us call such a system a **projective Steiner system** and denote such a system $PS(n, q)$.

Transversals in projective spaces

Let P be a projective space. A set S of subspaces of P is called **skew** if no two distinct subspaces of S have a point in common. Let S be a set of skew subspaces. A line is called a **transversal** of S if it intersects each subspace of S in exactly one point.

Lemma 2 ([2, Lemma 2.4.1])

Let ℓ_1 and ℓ_2 be two skew lines from P , and denote by t a point outside ℓ_1 and ℓ_2 . Then there is at most one transversal of ℓ_1 and ℓ_2 through t . If P is 3-dimensional then there is exactly one transversal of ℓ_1 and ℓ_2 through t .

Let P be a 3-dimensional projective space. A nonempty set R of skew lines of P is called a **regulus** if the following statements are true:

- (1) Through each point of each line of R there is a transversal of R .
- (2) Through each point of a transversal of R there is a line of R .

It follows from the definition that the set R_{opp} of all transversals of a regulus R again form a regulus; we call it the **opposite regulus** of R .

If P is a 3-dimensional projective space over finite field $\mathbb{F}_q$ then any regulus consists of exactly $q + 1$ lines.

Theorem 2 ([2, Theorem 2.4.3])

Let P be a 3-dimensional projective space over field $\mathbb{F}$. Let ℓ_1, ℓ_2, ℓ_3 be three skew lines of P . Then there is exactly one regulus through ℓ_1, ℓ_2, ℓ_3 .

Reguli in $AG(3, q)$

Let us define the notion of a regulus in $AG(3, q)$.

Let A be an affine space. A set S of subspaces of A is called **skew** if no two distinct subspaces of S are parallel and no of them have a point in common. Let S be a set of skew subspaces. A line is called a **transversal** of S if it intersects each subspace of S in exactly one point.

For an affine space $AG(3, q)$, a nonempty set R of skew lines of $AG(n, q)$ is called a **regulus** if the following statements are true:

- (1) Through each point of each line of R there is a transversal of R .
- (2) Through each point of a transversal of R there is a line of R .

It follows from the definition that the set R_{opp} of all transversals of a regulus R again form a regulus; we call it the **opposite regulus** of R .

Note that any regulus in $AG(3, q)$ consists of exactly q lines.

An explicit example of a regulus in $AG(n, q)$

Let $\langle a \rangle$ denote the linear span of the vector a . Consider two triples of $AS(n, 3)$:

$$t_1 = \{0, a, 2a\} = \langle a \rangle$$

and

$$t_2 = \{0, b, 2b\} = \langle b \rangle.$$

Each of these triples is a subspace of $V(n, 3)$ of dimension 1. Let c be a point of $AS(n, 3)$ such that c is not a linear combination of points a and b . Note that there are $3^n - 9$ different choices for the point c . Consider four more triples of $AS(n, 3)$:

$$t_3 = \{b, c + a + b, 2c + 2a + b\} = \langle c + a \rangle + b,$$

$$t_4 = \{a, c + b + a, 2c + 2b + a\} = \langle c + b \rangle + a,$$

$$t_5 = \{2b, 2c + a + 2b, c + 2a + 2b\} = \langle 2c + a \rangle + 2b,$$

$$t_6 = \{2a, 2c + b + 2a, c + 2b + 2a\} = \langle 2c + b \rangle + 2a.$$

An explicit example of a regulus in $AG(n, q)$

Note that in the block graph of $AS(n, 3)$ these six triples induce a subgraph which is isomorphic to $K_{3,3}$: one part consists of t_1, t_3 and t_5 and the other part consists of t_2, t_4 and t_6 . Since t_1, t_3, t_5 are skew as well as t_2, t_4, t_6 , the sets $S_1 = \{t_1, t_3, t_5\}$ and $S_2 = \{t_2, t_4, t_6\}$ are reguli that are opposite to each other.

An explicit example of a regulus in $AG(n, q)$

Now let point c be a linear combination of points a and b . Without loss of generality, we set $c = 0$. In this case

$$t_1 = \{0, a, 2a\} = \langle a \rangle,$$

$$t_2 = \{0, b, 2b\} = \langle b \rangle,$$

$$t_3 = \{b, a + b, 2a + b\} = \langle a \rangle + b,$$

$$t_4 = \{a, b + a, 2b + a\} = \langle b \rangle + a,$$

$$t_5 = \{2b, a + 2b, 2a + 2b\} = \langle a \rangle + 2b,$$

$$t_6 = \{2a, b + 2a, 2b + 2a\} = \langle b \rangle + 2a$$

and these six triples also induce a subgraph which is isomorphic to $K_{3,3}$. Note that in this case all triples lie in the same plane, which is isomorphic to $AG(2, 3)$ and since t_1, t_3, t_5 are parallel as well as t_2, t_4, t_6 , the sets $S_1 = \{t_1, t_3, t_5\}$ and $S_2 = \{t_2, t_4, t_6\}$ are two parallel classes in $AG(2, 3)$.

An explicit example of a regulus in $AG(n, q)$

Next, we describe a generalisation of the examples above for $AG(n, q)$.

Consider two families of lines in $AS(n, q)$:

$$S_1 = \{\langle kc + b \rangle + ka \mid k \in \mathbb{F}_q\} \text{ and } S_2 = \{\langle kc + a \rangle + kb \mid k \in \mathbb{F}_q\},$$

where a, b and c are three linearly independent points.

Remark 1

If we let point c be a linear combination of points a and b (without loss of generality, let $c = 0$), then all lines from $S_1 \cup S_2$ lie in the same plane, which is isomorphic to $AG(2, q)$. Moreover, lines from S_1 and lines from S_2 form two different parallel classes in that space.

An explicit example of a regulus in $AG(n, q)$

Proposition 1

The sets of lines S_1 and S_2 are a pair of opposite reguli in the 3-dimensional subspace $\langle a, b, c \rangle$ of $V(n, q)$ (a 3-dimensional subspace in $AG(n, q)$).

Proposition 2

Let R and R_{opp} be a regulus in $PG(3, q)$ and its opposite regulus, respectively. Let $\ell \in R$ and $\ell_{opp} \in R_{opp}$ be two arbitrary lines from the reguli, and let H be the plane determined by the intersecting lines ℓ and ℓ_{opp} . Then the removal of the points of H from $PG(3, q)$ leaves an affine geometry $AG(3, q)$ with a regulus R' and opposite regulus $(R_{opp})'$, where R' and $(R_{opp})'$ are obtained from R and R_{opp} by removal the lines ℓ and ℓ_{opp} together with their points.

Extension of special affine reguli

Proposition 3

Let R and R_{opp} be a pair opposite reguli in $AG(3, q)$ such that the systems of the pairwise skew lines from R and the pairwise skew lines from R_{opp} can be embedded into two classes of parallel planes, respectively. Then the pair of opposite reguli R and R_{opp} can be extended to a pair of opposite reguli in the projective closure of $AG(3, q)$ (isomorphic to $PG(3, q)$).

Affine reguli

Proposition 4

The pair of opposite reguli S_1 and S_2 in the three-dimensional affine space $\langle a, b, c \rangle$ from Proposition 1 can be embedded into two classes of parallel planes in $\langle a, b, c \rangle$.

Corollary 1

The pair of opposite reguli S_1 and S_2 in the three-dimensional affine space $\langle a, b, c \rangle$ from Proposition 1 can be extended to a pair of opposite reguli in the projective closure of $\langle a, b, c \rangle$ (isomorphic to $PG(3, q)$).

Problem 1

Is there a regulus in $AG(3, q)$ whose lines cannot be embedded into a class of parallel planes?

Eigenfunctions

Let θ be an eigenvalue of a graph G . A real-valued function f on the vertex set of G is called an **eigenfunction** of the graph G corresponding to the eigenvalue θ (or a **θ -eigenfunction** of G), if $f \not\equiv 0$ and for any vertex u in G the condition

$$\theta \cdot f(u) = \sum_{w \in G(u)} f(w) \quad (1)$$

holds, where $G(u)$ is the set of neighbours of the vertex u .

The summary of results on the problem of finding the minimum cardinality of support of eigenfunctions of graphs and characterising the optimal eigenfunctions is presented in [3].

[3] E. Sotnikova, A. Valyuzhenich, *Minimum supports of eigenfunctions of graphs: a survey*, Art Discrete Appl. Math. **4** (2021), no. 2, Paper No. 2.09, 34 pp.

Eigenfunctions of strongly regular graphs

Lemma 3 ([4, Corollary 1])

Let G be a primitive strongly regular graph with parameters (v, k, λ, μ) and let θ be a non-principal eigenvalue of G . Then an eigenfunction of G corresponding to the eigenvalue θ has at least

$$1 + |\theta| + \left\lceil \frac{(\theta - \lambda)\theta - k}{\mu} \right\rceil$$

non-zeros.

The lower bound given in Lemma 3 is known as the **weight-distribution bound**.

[4] D. S. Krotov, I. Yu. Mogilnykh, V. N. Potapov, *To the theory of q -ary Steiner and other-type trades*, Discrete Math. **339** (2016), no. 3, p. 1150–1157

Eigenfunctions of strongly regular graphs

Corollary 2 ([5, Corollary 2.3])

Let G be a primitive strongly regular graph with non-principal eigenvalues $s < 0 < r$. Then an eigenfunction of G corresponding to the eigenvalue r has at least $2(r + 1)$ non-zeroes, and an eigenfunction corresponding to the eigenvalue s has at least $-2s$ non-zeroes.

[5] S. Goryainov, L. Shalaginov, C. H. Yip, *On eigenfunctions and maximal cliques of generalised Paley graphs of square order* (2022), <https://arxiv.org/abs/2203.16081>

Eigenfunctions of the block graphs

Problem 2

Characterise the Steiner systems $S(2, m, n)$ for the block graphs of which the weight-distribution bound is tight for the negative eigenvalue s .

Eigenfunctions of the block graphs

Proposition 5

Let $S(2, m, n)$ be a Steiner system in which there exist two intersecting blocks contained in a subsystem isomorphic to the affine plane $AG(2, m)$. Then for the block graph of such a Steiner system there exists an $(-m)$ -eigenfunction with $2m$ non-zeros. In particular, this means that the weight-distribution bound is tight for the eigenvalue $-m$ of the block graph of such Steiner system $S(2, m, n)$.

Remark 2

Natural examples of Steiner systems satisfying the condition of Proposition 5 are $AS(n, q)$, where $n \geq 3$ and q is a prime power, and Hall triple systems (that is Steiner triple systems in which any three non-collinear points generate an affine plane $AG(2, 3)$).

Eigenfunctions of the block graphs

Proposition 6

Let $S(2, q + 1, n)$ be a Steiner system in which there exist a subsystem isomorphic to the projective space $PG(3, q)$. Let R be a regulus contained in $PG(3, q)$ and R_{opp} be its opposite regulus, $|R| = |R_{opp}| = q + 1$. Then the function f defined on the vertices of the block graph of $S(2, q + 1, n)$ as

$$f(x) = \begin{cases} 1 & \text{if } x \in R \\ -1 & \text{if } x \in R_{opp} \\ 0 & \text{otherwise} \end{cases}$$

is a $-(q+1)$ -eigenfunction of the block graph of $S(2, q+1, n)$. In particular, this means that the weight-distribution bound is tight for the eigenvalue $-q$ of the block graph of such Steiner system $S(2, q + 1, n)$.

Eigenfunctions of the block graphs

Remark 3

Natural examples of Steiner systems satisfying the condition of Proposition 6 are $PS(n, q)$, where $n \geq 3$ and q is a prime power. Note that the tightness of the weight-distribution bound for the smallest eigenvalue of Grassmann graphs was shown in general in [3, Theorem 5].

[4] D. S. Krotov, I. Yu. Mogilnykh, V. N. Potapov, *To the theory of q -ary Steiner and other-type trades*, Discrete Math. **339** (2016), no. 3, p. 1150–1157

Eigenfunctions of the block graphs

Proposition 7

Let $S(2, q, n)$ be a Steiner system in which there exists a subsystem isomorphic to the affine space $AG(3, q)$. Then for the block graph of such Steiner system there exists an $(-q)$ -eigenfunction with $2q$ non-zeros. In particular, this means that the weight-distribution bound is tight for the eigenvalue $-q$ of the block graph of such Steiner system $S(2, q, n)$.

Remark 4

Natural examples of Steiner systems satisfying the condition of Proposition 7 are $AS(n, q)$, where $n \geq 3$ and q is a prime power.

Eigenfunctions of the block graphs

Let R be a regulus contained in $PG(3, q)$ and R_{opp} be its opposite regulus, $|R| = |R_{opp}| = q + 1$. Consider a plane H , which does not contain the lines from R and R_{opp} , but contains $q + 1$ points from R . The removal of the points of H from $PG(3, q)$ leaves an affine geometry $AG(3, q)$ with a two sets of $q + 1$ lines R' and $(R_{opp})'$, where R' and $(R_{opp})'$ are obtained from R and R_{opp} by removal of $q + 1$ points.

Eigenfunctions of the block graphs

Proposition 8

Let $S(2, q, n)$ be a Steiner system in which there exists a subsystem isomorphic to the affine space $AG(3, q)$. Then the function f defined on the vertices of the block graph of $S(2, q, n)$ as

$$f(x) = \begin{cases} 1 & \text{if } x \in R' \\ -1 & \text{if } x \in (R_{opp})' \\ 0 & \text{otherwise} \end{cases}$$

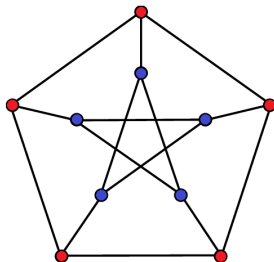
is a $(-q)$ -eigenfunction of the block graph of $S(2, q, n)$ with $2(q+1)$ non-zeros.

Equitable partitions

Let $\Pi = V_1 \cup V_2 \cup \dots \cup V_t$ be the partition of the vertex set of a graph G into t parts with the property that every vertex of V_i has exactly p_{ij} neighbours in V_j .

Then Π is called an **equitable** t -partition of G .

Matrix $P_\Pi = (p_{ij})_{t \times t}$ is called the **quotient matrix** of equitable partition.



$$P = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$$

SRGs with eigenvalue $s = -3$

Lemma 4 ([6, Theorem 5.1])

Primitive strongly regular graphs with the negative eigenvalue $s = -3$ are:

- (1) Latin-square graphs;*
- (2) Block graphs of Steiner triple systems $STS(n)$;*
- (3) Finitely many other graphs.*

[6] A. Neumaier, *Strongly regular graphs with smallest eigenvalue $-m$* , Arch. Math. **33** (1979), p. 392–400.

Equitable 2-partitions of graphs with eigenvalue -3

- Equitable 2-partitions of the Johnson graphs $J(n, 3)$ (2012–2022, [7, 8])
- Equitable 2-partitions of the bilinear-forms graph $Bil_2(2 \times d)$ (2017, [9])
- Equitable 2-partitions of Latin-square graphs (2019, [10])

[7] A. L. Gavriluk, S. V. Goryainov, *On perfect 2-colorings of Johnson graphs $J(v, 3)$* , Journal of Combinatorial Designs **21** (2013) no. 6, p. 232–252.

[8] R. J. Evans, A. L. Gavriluk, S. V. Goryainov, K. Vorob'ev *Equitable 2-partitions of the Johnson graphs $J(n, 3)$* (2022), <https://arxiv.org/abs/2206.15341>

[9] S. V. Goryainov, *Perfect 2-colourings of bilinear forms graph $Bil_2(2 \times d)$* , Talk at the International Workshop on Bannai-Ito Theory (2017).

[10] R. A. Bailey, P. J. Cameron, A. L. Gavriluk, S. V. Goryainov, *Equitable partitions of Latin-square graphs*, Journal of Combinatorial Designs **27** (2019) no. 3, p. 142–160.

Eigenvalues of quotient matrices

Let P be the quotient matrix of a t -partition of a graph G . Then each eigenvalue of P is an eigenvalue of the adjacency matrix of G (see [11, Theorem 9.3.3]).

Let G be a k -regular graph. If $t = 2$, then exactly one non-principal eigenvalue θ of G is an eigenvalue of P . In this case we say that the equitable 2-partition is **θ -equitable**.

Lemma 5

Let Π be a θ -equitable 2-partition of k -regular graph G with quotient matrix

$$P_{\Pi} = \begin{pmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{pmatrix}.$$

Then the eigenvalues of P_{Π} are given by $k = p_{11} + p_{12} = p_{21} + p_{22}$ and $\theta = p_{11} - p_{21} = p_{22} - p_{12}$.

Equitable 2-partitions as eigenfunctions

Let $\Pi = V_1 \cup V_2$ be a 2-partition of the vertex set of a k -regular graph G . For any $a, b \in \mathbb{R}$, consider a function $f : V(G) \rightarrow \mathbb{R}$ such that for any $u \in V(G)$

$$f(u) = \begin{cases} a, & \text{if } u \in V_1 \\ b, & \text{if } u \in V_2 \end{cases}.$$

Let θ be a non-principal eigenvalue of G .

Lemma 6

The following statements are equivalent.

- (1) The partition Π is an equitable 2-partition of G corresponding to the eigenvalue θ , and $\begin{pmatrix} a \\ b \end{pmatrix}$ is an eigenvector of the quotient matrix P corresponding to the eigenvalue θ .*
- (2) The function f is an eigenfunction of G corresponding to the eigenvalue θ .*

Equitable 2-partitions as eigenfunctions

In view of Lemma 6, the investigation of equitable 2-partitions is equivalent to the investigation of eigenfunctions with two values. Since eigenfunctions corresponding to distinct eigenvalues of a graph are orthogonal, we can use θ_1 -eigenfunctions of a regular graph G to restrict possible θ_2 -equitable 2-partitions of the graph G whenever $\theta_1 \neq \theta_2$.

Equitable 2-partitions as eigenfunctions

For a non-principal eigenvalue θ_1 , let f_1 be a $(1, -1, 0)$ -valued θ_1 -eigenfunction of G . Denote by $Supp^+(f_1)$ and $Supp^-(f_1)$ the sets of vertices of G where f_1 takes values 1 and -1 , respectively. Thus, we have an equitable partition, defined by preimages of 1, -1 and 0

$$\Pi = Supp^+(f_1) \cup Supp^-(f_1) \cup V(G) \setminus (Supp^+(f_1) \cup Supp^-(f_1)).$$

Since f_1 is orthogonal to the (1) -valued function, which is an k -eigenfunction of G , we have $|Supp^+(f_1)| = |Supp^-(f_1)|$.

Equitable 2-partitions as eigenfunctions

Take a non-principal eigenvalue θ_2 of G such that $\theta_2 \neq \theta_1$. Consider an arbitrary θ_2 -equitable 2-partition $\Pi = V_1 \cup V_2$ of G and consider (a, b) -valued θ_2 -eigenfunction f_2 of G representing Π (see Lemma 6).

Lemma 7

The sets $\text{Supp}^+(f_1)$ and $\text{Supp}^-(f_1)$ have the same number of vertices from V_1 .

Cameron-Liebler line classes

In [12, Definition 2.1], among other equivalent definitions, a **Cameron-Liebler line class** $\mathcal{L}$ in $PG(3, q)$ is defined to be a class of lines such that the condition $|R \cap \mathcal{L}| = |R_{opp} \cap \mathcal{L}|$ holds for every regulus R and its opposite R_{opp} .

But in view of Lemma 7 and Proposition 6, the condition $|R \cap V_1| = |R_{opp} \cap V_1|$ holds for any part V_1 of an equitable 2-partition of the block graph of projective Steiner system $PS(3, q)$ (the Grassmann graph $J_q(4, 2)$) corresponding to its positive non-principal eigenvalue. Thus, Cameron-Liebler line classes in $PG(3, q)$ correspond to equitable 2-partitions of the Grassmann graph $J_q(4, 2)$ corresponding to its positive non-principal eigenvalue.

[12] M. Rodgers, *On some new examples of Cameron-Liebler line classes*, PhD thesis, University of Colorado at Denver, 2012, <https://cage.ugent.be/~mrodders/papers/thesis.pdf>

Cameron-Liebler line classes

Moreover, Cameron-Liebler line classes correspond to equitable 2-partitions of the block graphs of $AS(n, q)$ and $PS(n, q)$ for the positive non-principal eigenvalue (see [13]).

Problem 3

Characterise Cameron-Liebler line classes of $AS(n, q)$ and $PS(n, q)$ (equitable 2-partitions of the block graphs for the positive non-principal eigenvalue).

Further, we plan to study more $(1, -1, 0)$ -valued $(-q)$ -eigenfunctions and $-(q+1)$ -eigenfunctions of the block graphs of $AS(n, q)$ and $PS(n, q)$, respectively, to get new restrictions on the Cameron-Liebler line classes.

Thanks for your attention!